

STRING FUNCTION

UPPERCASE

```
m1 x
Limit to 1000 rows

1 • CREATE TABLE STUDENT (
2     Student_ID INT,
3     Student_Name VARCHAR(15),
4     Class VARCHAR(8),
5     Marks INT,
6     Fee_Paid DECIMAL(10,2),
7     Date_of_Birth DATE,
8     Admission_Date DATE
9 );
10 • INSERT INTO STUDENT
11     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12     VALUES
13     (201, 'Rahul', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14     (202, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15     (203, 'Kiran', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16     (204, 'Rohan', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17     (205, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21     UPPER(Student_Name) AS Uppercase_Name
22 FROM STUDENT;
```

OUTPUT

	Student_Name	Uppercase_Name
▶	Rahul	RAHUL
	Arjun	ARJUN
	Kiran	KIRAN
	Rohan	ROHAN
	Vivek	VIVEK

LOWERCASE

```
m1 x
Limit to 1000 rows

1 • CREATE TABLE STUDENT (
2     Student_ID INT,
3     Student_Name VARCHAR(15),
4     Class VARCHAR(8),
5     Marks INT,
6     Fee_Paid DECIMAL(10,2),
7     Date_of_Birth DATE,
8     Admission_Date DATE
9 );
10 • INSERT INTO STUDENT
11     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12     VALUES
13     (201, 'Rahul', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14     (202, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15     (203, 'Kiran', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16     (204, 'Rohan', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17     (205, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21     LOWER(Student_Name) AS Lowercase_Name
22 FROM STUDENT;
```

OUTPUT

	Student_Name	Lowercase_Name
►	Rahul	rahul
	Arjun	arjun
	Kiran	kiran
	Rohan	rohan
	Vivek	vivek

LENGTH

```
m1 x
Limit to 1000 rows

4      Student_Name VARCHAR(15),
5      Class VARCHAR(8),
6      Marks INT,
7      Fee_Paid DECIMAL(10,2),
8      Date_of_Birth DATE,
9      Admission_Date DATE
10    );
11 • INSERT INTO STUDENT
12      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
13    VALUES
14      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
15      (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
16      (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
17      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
18      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
19 • SELECT *
20 FROM STUDENT;
21 • SELECT Student_Name,
22      LEN(Student_Name) AS Name_Length
23 FROM STUDENT;
```

OUTPUT

	Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
▶	115	Mohith Gowda MR	10A	85	55000.00	2007-02-13	2023-06-10
	116	Rahul	10A	92	53000.00	2007-11-20	2023-06-12
	117	Arjun	10B	78	45000.00	2007-08-08	2023-06-11
	118	Karthik	10B	88	46000.00	2007-01-03	2023-06-15
	119	Vivek	10A	65	49000.00	2007-01-30	2023-06-13

STUDENT 5

CONCAT

```
m1
5      Student_Name VARCHAR(15),
6      Class VARCHAR(8),
7      Marks INT,
8      Fee_Paid DECIMAL(10,2),
9      Date_of_Birth DATE,
10     Admission_Date DATE
11 );
12
13 • INSERT INTO STUDENT
14     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
15 VALUES
16     (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
17     (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
18     (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
19     (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
20     (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
21
22 • SELECT *
23 FROM STUDENT;
24
25 • SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
26 FROM STUDENT;
```

OUTPUT

	Student_Details
►	Mohith Gowda MR - Class 10A
	Rahul - Class 10A
	Arjun - Class 10B
	Karthik - Class 10B
	Vivek - Class 10A

SUBSTRING

```
m1
6      Class VARCHAR(8),
7      Marks INT,
8      Fee_Paid DECIMAL(10,2),
9      Date_of_Birth DATE,
10     Admission_Date DATE
11 );
12
13 INSERT INTO STUDENT
14     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
15 VALUES
16     (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
17     (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
18     (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
19     (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
20     (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
21
22 SELECT *
23 FROM STUDENT;
24
25 SELECT Student_Name,
26        SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
27 FROM STUDENT;
```

OUTPUT

	Student_Name	First_Five_Characters
►	Mohith Gowda MR	Mohit
	Rahul	Rahul
	Arjun	Arjun
	Karthik	Karth
	Vivek	Vivek

NUMERIC FUNCTION

ROUND

```
m1 x
Limit to 1000 rows
3      Student_ID INT,
4      Student_Name VARCHAR(20),
5      Class VARCHAR(8),
6      Marks INT,
7      Fee_Paid DECIMAL(10,2),
8      Date_of_Birth DATE,
9      Admission_Date DATE
10 );
11 • INSERT INTO STUDENT
12     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
13     VALUES
14     (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
15     (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
16     (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
17     (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
18     (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
19 • SELECT *
20 FROM STUDENT;
21 • SELECT Student_Name,
22         Marks,
23         ROUND(Marks * 1.05, 0) AS Bonus_Marks
24 FROM STUDENT;
```

OUTPUT

	Student_Name	Marks	Bonus_Marks
►	Mohith Gowda MR	85	89
	Rahul	92	97
	Arjun	78	82
	Karthik	88	92
	Vivek	65	68

CEIL

```

m1
Limit to 1000 rows
1 • DROP TABLE IF EXISTS STUDENT;
2 • CREATE TABLE STUDENT (
3     Student_ID INT,
4     Student_Name VARCHAR(20),
5     Class VARCHAR(8),
6     Marks INT,
7     Fee_Paid DECIMAL(10,2),
8     Date_of_Birth DATE,
9     Admission_Date DATE
10 );
11 • INSERT INTO STUDENT
12     (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
13     VALUES
14     (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
15     (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
16     (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
17     (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
18     (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
19 • SELECT *
20 FROM STUDENT;
21 • SELECT Student_Name,
22     Marks / 10.0 AS Marks_Value,

```

OUTPUT

	Student_Name	Marks_Value	Rounded_Up
►	Mohith Gowda MR	8.5000	9
	Rahul	9.2000	10
	Arjun	7.8000	8
	Karthik	8.8000	9
	Vivek	6.5000	7

FLOOR

```

m1
Limit to 1000 rows

3      Student_ID INT,
4      Student_Name VARCHAR(20),
5      Class VARCHAR(8),
6      Marks INT,
7      Fee_Paid DECIMAL(10,2),
8      Date_of_Birth DATE,
9      Admission_Date DATE
10    );
11 •   INSERT INTO STUDENT
12      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
13    VALUES
14      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
15      (116, 'Rahul', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
16      (117, 'Arjun', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
17      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
18      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
19 •   SELECT *
20   FROM STUDENT;
21 •   SELECT Student_Name,
22          Marks / 10.0 AS Marks_Value,
23          FLOOR(Marks / 10.0) AS Rounded_Down
24   FROM STUDENT;

```

OUTPUT

	Student_Name	Marks_Value	Rounded_Down
►	Mohith Gowda MR	8.5000	8
	Rahul	9.2000	9
	Arjun	7.8000	7
	Karthik	8.8000	8
	Vivek	6.5000	6

MOD

```
m1* x
Limit to 1000 rows
2      Student_ID INT,
3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21      Marks,
22      Marks % 10 AS Remainder
23 FROM STUDENT;
```

OUTPUT

	Student_Name	Marks	Remainder
►	Mohith Gowda MR	85	5
	Rahul	92	2
	Arjun	78	8
	Karthik	88	8
	Vivek	65	5

ABS

```
m1 x
Limit to 1000 rows

2      Student_ID INT,
3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21      Marks,
22      ABS(Marks - 80) AS Difference_From_80
23 FROM STUDENT;
```

OUTPUT

	Student_Name	Marks	Difference_From_80
►	Mohith Gowda MR	85	5
	Rahul	92	12
	Arjun	78	2
	Karthik	88	8
	Vivek	65	15

DATE FUNCTIONS:

Extracting the year

```
m1* x
Limit to 1000 rows
3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21      Date_of_Birth,
22      YEAR(Date_of_Birth) AS Birth_Year
23 FROM STUDENT;
24
```

OUTPUT

	Student_Name	Date_of_Birth	Birth_Year
►	Mohith Gowda MR	2007-02-13	2007
	Rahul	2007-11-20	2007
	Arjun	2007-08-08	2007
	Karthik	2007-01-03	2007
	Vivek	2007-01-30	2007

Extracting the month

```

m1
Limit to 1000 rows

2      Student_ID INT,
3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21      Date_of_Birth,
22      MONTH(Date_of_Birth) AS Birth_Month
23 FROM STUDENT;

```

OUTPUT

	Student_Name	Date_of_Birth	Birth_Month
►	Mohith Gowda MR	2007-02-13	2
	Rahul	2007-11-20	11
	Arjun	2007-08-08	8
	Karthik	2007-01-03	1
	Vivek	2007-01-30	1

Extracting the day

```

m1
Limit to 1000 rows

2      Student_ID INT,
3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2007-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-01-30', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT Student_Name,
21      Date_of_Birth,
22      DAY(Date_of_Birth) AS Birth_Day
23 FROM STUDENT;

```

OUTPUT

	Student_Name	Date_of_Birth	Birth_Day
►	Mohith Gowda MR	2007-02-13	13
	Rahul	2007-11-20	20
	Arjun	2007-08-08	8
	Karthik	2007-01-03	3
	Vivek	2007-01-30	30

Calculating current age

```
m1* x
Limit to 1000 rows

3      Student_Name VARCHAR(20),
4      Class VARCHAR(8),
5      Marks INT,
6      Fee_Paid DECIMAL(10,2),
7      Date_of_Birth DATE,
8      Admission_Date DATE
9  );
10 • INSERT INTO STUDENT
11      (Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
12  VALUES
13      (115, 'Mohith Gowda MR', '10A', 85, 55000.00, '2007-02-13', '2023-06-10'),
14      (116, 'Arjun', '10A', 92, 53000.00, '2007-11-20', '2023-06-12'),
15      (117, 'Rahul', '10B', 78, 45000.00, '2007-08-08', '2023-06-11'),
16      (118, 'Karthik', '10B', 88, 46000.00, '2008-01-03', '2023-06-15'),
17      (119, 'Vivek', '10A', 65, 49000.00, '2007-12-12', '2023-06-13');
18 • SELECT *
19 FROM STUDENT;
20 • SELECT
21      Student_Name,
22      Date_of_Birth,
23      DATEDIFF(YEAR, Date_of_Birth, GETDATE()) AS Age
24 FROM STUDENT;
```

OUTPUT

	Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
▶	115	Mohith Gowda MR	10A	85	55000.00	2007-02-13	2023-06-10
	116	Rahul	10A	92	53000.00	2007-11-20	2023-06-12
	117	Arjun	10B	78	45000.00	2007-08-08	2023-06-11
	118	Karthik	10B	88	46000.00	2007-01-03	2023-06-15
	119	Vivek	10A	65	49000.00	2007-01-30	2023-06-13

AGGREGATE FUNCTION GROUP BY

```
m1
Limit to 1000 rows

2      Student_ID INT,
3      Name VARCHAR(50),
4      Class INT,
5      Marks INT
6  );

7
8  INSERT INTO STUDENT (Student_ID, Name, Class, Marks)
9  VALUES
10     (1, 'Mohith Gowda MR', 1, 85),
11     (2, 'Rahul', 1, 65),
12     (3, 'Arjun', 1, 75),
13     (4, 'Karthik', 2, 80),
14     (5, 'Vivek', 2, 65);
15
16 • SELECT *
17   FROM STUDENT;
18
19 • SELECT
20     Class,
21     COUNT(*) AS Total_Students,
22     AVG(Marks) AS Average_Marks
23   FROM STUDENT
```

OUTPUT

	Class	Total_Students	Average_Marks
▶	10A	81	80.6667
	10B	54	83.0000

ORDER BY

```
m1 x
Limit to 1000 rows

1 CREATE TABLE STUDENT (
2     Student_ID INT,
3     Name VARCHAR(50),
4     Class INT,
5     Marks INT
6 );
7
8 INSERT INTO STUDENT (Student_ID, Name, Class, Marks)
9 VALUES
10     (1, 'Mohith Gowda MR', 1, 85),
11     (2, 'Rahul', 1, 65),
12     (3, 'Arjun', 1, 75),
13     (4, 'Karthik', 2, 80),
14     (5, 'Vivek', 2, 65);
15
16 SELECT *
17 FROM STUDENT;
18
19 SELECT *
20 FROM STUDENT
21 ORDER BY Name ASC;
```

OUTPUT

	Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
▶	115	Mohith Gowda MR	10A	85	55000.00	2007-02-13	2023-06-10
	116	Rahul	10A	92	53000.00	2007-11-20	2023-06-12
	117	Arjun	10B	78	45000.00	2007-08-08	2023-06-11
	118	Karthik	10B	88	46000.00	2007-01-03	2023-06-15
	119	Vivek	10A	65	49000.00	2007-01-30	2023-06-13